# Acute First-Time Hamstring Strains During High-Speed Running

## A Longitudinal Study Including Clinical and Magnetic Resonance Imaging Findings

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**Background:** Hamstring muscle strain is one of the most common injuries in sports. Still, knowledge is limited about the progression of clinical and magnetic resonance imaging characteristics and their association with recovery time in athletes.

**Hypothesis:** Knowing the anatomical location and extent of an acute first-time hamstring strain in athletes is critical for the prognosis of recovery time.

**Study Design:** Case series (prognosis); Level of evidence, 2.

**Methods:** Eighteen elite sprinters with acute first-time hamstring strains were prospectively included in the study. All subjects were examined, clinically and with magnetic resonance imaging, on 4 occasions after injury: at day 2 to 4, 10, 21, and 42. The clinical follow-up period was 2 years.

**Results:** All sprinters were injured during competitive sprinting, and the primary injuries were all located in the long head of the biceps femoris muscle. There was an association between the time to return to pre-injury level (median, 16; range, 6-50 weeks) and the extent of the injury, as indicated by the magnetic resonance imaging parameters. Involvement of the proximal free tendon, as estimated by MRI, and proximity to the ischial tuberosity, as estimated both by palpation and magnetic resonance imaging, were associated with longer time to return to pre-injury level.

**Conclusion:** Careful palpation during the first 3 weeks after injury and magnetic resonance imaging investigation performed during the first 6 weeks after injury provide valuable information that can be used to predict the time to return to pre-injury level of performance in elite sprinting.

**Keywords:** magnetic resonance imaging; palpation; recovery; rehabilitation

The high prevalence of hamstring strains is well documented in sports, for example, Australian football,[14] rugby,[3] and soccer.[20] Attempts to sort out specific injury situations in these often complex sports have pointed to high-speed running as a particularly injury-prone activity.[3,6,18,20] However, until now, no study has been performed exclusively on sprinters. Results from such a homogeneous group, exposed to a specific risk activity, for example, undisturbed, straightforward, high-speed running, may potentially provide leads as to the underlying mechanisms, should the characteristics of the injury be distinct and consistent.

Hamstring injuries are often detrimental for the athlete, causing prolonged absence from competition and high rates of injury recurrence.[3,13] Establishing the normal progression of the injury for a certain category of athletes is important both in relation to prescribing appropriate rehabilitation and predicting recovery time. In this context, it is critical to establish valid criteria for complete healing to avoid premature return to full activity and the concomitant increased risk of re-injury. In an earlier study we showed that recovery of knee flexor strength and flexibility in hip flexion to more than 90%

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of that of the uninjured leg does not imply that a sprinter considers himself or herself ready to return to full activity.[1] In a recent study on hamstring injuries in Australian football players,[4] remaining signs of injury were observed on magnetic resonance imaging (MRI) 6 weeks after injury in 36% of the players with an MRI-verified hamstring injury at baseline, although all but 1 player had returned to play at that time. Adding to the complexity of the issue is the possibility that the criteria for recommending return to full activity may vary between athletic activities.

Making accurate predictions of recovery time is essential with all injuries, particularly in sports. Earlier studies have indicated that the injury location in terms of specific muscle in the hamstring muscle synergy,[4] the region in the tendon-muscle complex,[2] as well as size of the injury[4,8,15,17] may constitute indicators of time back to pre-injury level. Detailed information about the injury characteristics can be obtained with imaging techniques, such as MRI; however, simpler clinical techniques would be preferable from a cost and availability point of view. Recent studies have indicated that results of clinical tests may be as useful in predicting rehabilitation time as MRI data, at least in the case of minor and moderate hamstring injuries.[16] The prognostic value of defining the location of the injury with palpation has not been systematically evaluated. However, a group of dancers with a more cranial location of the point of highest palpation pain than the sprinters group was found to demand longer rehabilitation time.[1]

The primary aims of this study were to investigate acute, first-time hamstring strains in sprinters with respect to the occurrence and progression of both clinical and MRI signs of injury during the first 6 weeks after injury, as well as the extent of correlation between clinical and MRI findings and their association with time to return to pre-injury level.

## METHODS

### Subjects

Eighteen sprinters (8 female and 10 male; age range, 15-28 years; body mass, 53-84 kg; height, 1.59-1.82 m) competing on a national or international level were included. Approval of the study was obtained from the Ethical Committee of the Karolinska Institutet.

### Recruitment Procedure

The subjects were recruited in response to information given to all major track and field clubs in Sweden by the Swedish Athletic Association. All sprinters who encountered an acute injury to their rear thigh were asked to contact the test leader (C.A.) by phone on the day of injury. They were then informed about the background of the study and invited to participate on a completely voluntary basis. During this initial phone contact, they also received information on how to deal initially with the injury, for example, not to use NSAIDs until they met with the test leader. The first 48 hours involved strict adherence to the RICE (Rest, Ice, Compression, and Elevation) principles[10] in an effort to minimize both bleeding and edema.

### Inclusion/Exclusion Criteria

To be included, the sprinters had to have a first-time, acute, sudden pain in the posterior thigh when training or competing. The initial clinical examination, 2 days after injury, had to reveal localized pain when palpating the hamstring muscles, local pain when performing a passive straight leg raise test, and an increased pain when adding an isometric hamstring contraction during that test. Exclusion criteria included verified, or even suspected, earlier hamstring strains in the same leg, extrinsic trauma to the posterior thigh, ongoing or chronic low back problems, and pregnancy. The recruitment time lasted 3 years. In all, 24 sprinters had to be excluded, all due to an earlier hamstring injury. The ensuing MRI investigation verified a hamstring injury in all sprinters included in the study.

### Clinical Examinations

The subjects were clinically examined on 4 occasions: 2, 10, 21, and 42 days after the acute injury. With the exception of the first examination, all subsequent clinical examinations were performed on the same day as the MRI investigations. At the first visit, the sprinters were interviewed about the injury situation, that is, the movements or exercises during which the injury occurred. Presence of pre-injury symptoms from the posterior thigh, for example, muscle pain or staleness, was registered. Inspection for ecchymosis was performed. The functional observations encompassed ability to walk without crutches and walk on level ground without pain. Measurements of isometric knee flexor strength and range of motion in hip flexion with straight knee were also included in the test battery.[1]

Palpation of the rear thigh was performed with the subjects prone and the knee extended. First, the origin of the hamstring muscles on the ischial tuberosity was identified. The subjects were then asked to activate their hamstrings by performing an isometric contraction with resistance against the heel and no flexion of the knee allowed. The palpation started approximately 5 cm cranial to the hamstring origin and continued without interruption to the respective muscle insertions on the lower leg and then repeated in the other direction. The point where the subject noted the highest pain on palpation was marked on the skin and the distance between this point and the palpated ischial tuberosity was measured. Also, the total length (cranial-to-caudal border) of the painful area on palpation was measured, and it was noted whether the painful area was situated in the medial or lateral part of the hamstring muscle complex. Each sprinter's self-estimated time to return to pre-injury level was registered (at 2 days after injury). Clinical examinations, functional observations, and interviews in all cases were performed by the same test leader (C.A.).

### MRI Investigations

The injured sprinters were to undergo 4 consecutive MRI investigations (MRI I-IV), at 4, 10, 21, and 42 days after the incident of acute hamstring strain. All MRI investigations were performed on a 1.0 Tesla superconductive MRI unit

TABLE 1
MRI Settings[a]

| Plane Sequence | TR/TE (ms) | TI (ms) | Averages | Matrix | FOV (mm) |
|---|---|---|---|---|---|
| Frontal STIR | 4000/30 | 150 | 2 | 294 × 512 | 420 × 480 |
| Sagittal STIR | 4000/30 | 150 | 2 | 210 × 512 | 300 × 480 |
| Transversal STIR | 5035/30 | 150 | 2 | 154 × 256 | 263 × 350 |
| Transversal T1 | 722/20 | | 1 | 265 × 512 | 263 × 350 |
| Transversal T2 | 5500/110 | | 1 | 168 × 256 | 219 × 350 |

[a]TR, time to repetition; TE, time to echo; TI, time to inversion; averages, number of excitations; matrix, resolution in number of pixels; FOV, field of view; STIR, short time inversion recovery.

(Magnetom Expert®, Siemens, Erlangen, Germany). The patients were positioned supine on a commercial phased array spine coil. First, sagittal and frontal short time inversion recovery (STIR) images were obtained with a large field of view from at least 5 cm above the ischial tuberosity to the knee. Then transversal, T1-weighted, T2-weighted, and STIR images were obtained covering the entire damaged area on the longitudinal images. Frontal and transversal images included also the uninjured side for comparisons at all 4 occasions. The thickness of the slices for all sequences was 5 mm, with a 0.5 mm gap. Details on the imaging settings are given in Table 1. The protocol was performed with a total examination time of approximately 50 min per subject at each occasion.

A muscle was considered injured when it contained high signal intensity, as compared with the uninjured side, on the STIR images. A tendon tissue was considered injured if it was thickened and/or had an intratendinous high signal and/or a collar of high signal intensity around it on the STIR images. If more than 1 muscle/tendon was injured, the one with greatest signal abnormality was considered the "primary" injury and the second largest, the "secondary" injury. In this study, MRI parameters (cf. below) were quantified only for the primary injury. Each injury was allocated to 1 or more of 6 different regions within the muscle-tendon complex (Figure 1).

The size of the primary injury was measured on the images as the maximal length (cranio-caudal extent), width (medio-lateral extent), and depth (antero-posterior extent). An attempt to estimate the volume of the injury was made by assuming that the injury had a shape of a rotational ellipsoid, that is, volume ≈ length × width × depth × 0.5.[4,17] The cross-sectional area of the injured muscle, as a percentage of the total cross-sectional area, was calculated at the level where the injury had the largest absolute cross-sectional distribution in the muscle.[4,17] The most cranial pole of the injury was identified and its cranio-caudal distance to the most caudal part of the ischial tuberosity was measured; hereafter this distance will be referred to as "distance to tuber."

The images of the first MRI investigation were independently analyzed by 2 experienced MRI radiologists who used the same evaluation protocol and were unaware of clinical details other than the suspected hamstring injury. Discordant cases (regarding either the presence or absence of hamstring muscle injury or the muscle or region involved) were reviewed and consensus reached. The subsequent quantitative MRI analyses were performed by 1 of the same radiologists (M.T.). At the first MRI investigation, a plain radiograph of the pelvis was obtained from all sprinters to detect possible skeletal injuries.

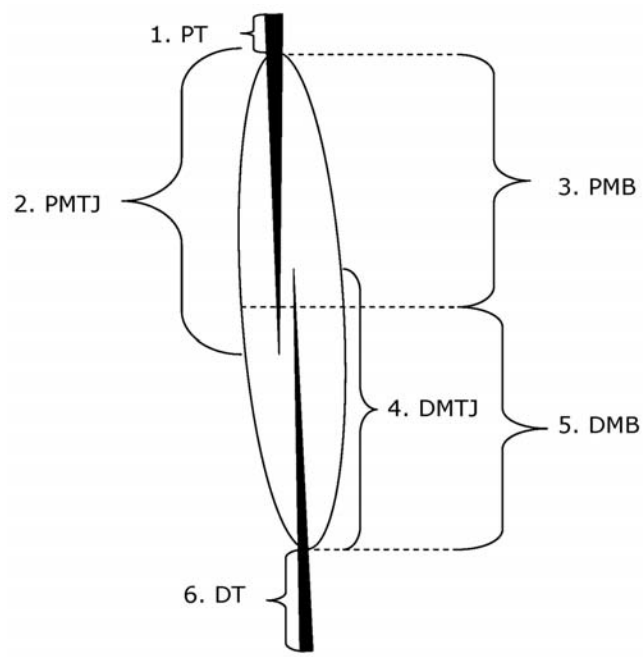


**Figure 1.** Schematic drawing in the frontal plane of the muscle-tendon complex of the long head of biceps femoris showing the 6 different regions used when analyzing the injury location and tissues involved: 1. proximal tendon (PT), ie, free tendon proximal to muscle fiber attachment; 2. proximal muscle-tendon junction (PMTJ), defined as the proximal intramuscular tendon and attached muscle fibers; 3. proximal muscle-belly (PMB), ie, muscle proximal to the midpoint of the whole muscle-belly; 4. distal muscle-tendon junction (DMTJ), defined as the distal intramuscular tendon and attached muscle fibers; 5. distal muscle-belly (DMB), ie, muscle distal to the midpoint of the whole muscle-belly; and 6. distal tendon (DT), ie, free tendon distal to muscle fiber attachment.

## Follow-up

All subjects received the same standardized, 3-part, progressive rehabilitation program, which was presented at the first,

second, and third clinical examination, respectively. The program included specific hamstring exercises for strength and flexibility as well as more complex whole-body exercises. Also, jogging with progressively increasing stride-length and variable pace running and interval running of gradually higher speed were included. A pain-free approach was used at all stages. After the last examination, an average of 6 weeks after injury, each athlete's own physician and/or physiotherapist administered continued rehabilitation.

The sprinters were asked to register the first week when they could train or compete unimpeded at their pre-injury level, that is, sprinting at similar best times as before the injury. If re-injury occurred in the same thigh, the sprinters were asked to immediately contact the test leader by phone. Furthermore, regular follow-up phone contacts were done by the test leader at 3, 12, and 24 months after the occurrence of the initial injury, and any symptoms or problems from the previously injured thigh were noted. The investigation was terminated when the entire follow-up period (24 months) was completed for all subjects.

### Statistics

Shapiro-Wilk's W test was applied to examine normality in the distribution of data. Repeated measures ANOVA with post-hoc Tukey test, Student *t* test, and Mann-Whitney *U* test were used to detect statistical differences. Pearson's rank order correlation was calculated between subjects' MRI parameters, palpation data, and time to return to pre-injury level. The significance level was set at $P < .05$, and tendencies were identified at $0.05 \leq P < .1$.

## RESULTS

### Clinical Examinations

All 18 sprinters reported that they had experienced their injuries during competition sprinting when the speed was maximal or close to maximal. They were all forced to stop running directly when the injury occurred, and 11 of the 18 (61%) actually fell. At the initial examination (2 days after injury), 15 sprinters (83%) were using crutches. None of the sprinters reported any pre-injury symptoms, for example, muscle stiffness or pain. All but 3 injuries occurred in the right leg.

On palpation of the injury, all sprinters noted their highest pain in the lateral part of the rear thigh. The mean distances (±1SD, range) from the point with the highest pain to the ischial tuberosity on the 4 test occasions were as follows: 12 (±6, 5-24) cm, 11 (±7, 3-24) cm, 12 (±7, 4-24) cm, and 11 (±8, 4-24) cm, respectively. The corresponding mean lengths (±1SD, range) of the painful area were as follows: 11 (±5, 5-24) cm, 7 (±3, 3-11) cm, 6 (±2, 4-9) cm, 5 (±2, 2-8), respectively. At the last occasion (6 weeks after injury), 6 subjects did not experience any pain on palpation and were therefore not included in these calculations. There was a significant correlation between the location of the point of highest pain during palpation and time back to pre-injury level at clinical examination I ($r = 0.695$, $P = .004$) and III ($r = 0.680$, $P = .005$) and a tendency toward a correlation at examination II ($r = 0.503$, $P = .057$), that is, the more cranial the location, the longer the time back to pre-injury level. No correlation was present between the palpated length of the painful area and time back to pre-injury level. No ecchymosis was observed on the posterior thigh at any of the clinical examinations.

TABLE 2
Mean (and Individual) Times to Return to Pre-Injury Level[a]

| Injured Regions | n | Time to Return, Weeks |
|---|---|---|
| PMTJ + PT | 5 | 32 (16, 20, 25, 49, 50) |
| PMTJ | 5 | 14.2 (6, 8, 10, 21, 26) |
| PMTJ + DMTJ | 1 | 28 |
| PMTJ + PT + DMTJ | 1 | 49 |
| DMB | 2 | 7.5 (7, 8) |
| DMTJ | 2 | 14 (12, 16) |
| DMTJ + DT | 2 | 7 (6, 8) |

[a]n, number of sprinters with injuries in the various regions of the musculo-tendinous system of the long head of the biceps femoris, as defined in Figure 1; PMTJ, proximal muscle-tendon junction; PT, proximal tendon; DMTJ, distal muscle tendon junction; DMB, distal muscle-belly; DT, distal tendon.

At the initial clinical examination (2 days after injury) the sprinters estimated that their time back to pre-injury level would be 4 weeks (median, range 2-12 w). However, the actual time back turned out to be significantly longer, the median value being 16 weeks (range, 6-50 w) (Table 2). At 6 weeks after injury all the sprinters could jog without pain, but only 2 (11%) were able to train and/or compete at their pre-injury level according to their own judgment. During the 2-year follow-up, 3 sprinters (17%) encountered re-injuries of their hamstrings (at 8, 9, and 20 months after their first-time injury, respectively) leading to a forced retirement from competitive sprinting for 2 of them. None of the sprinters had any other serious injuries during the 2-year follow-up period.

### MRI Investigations

The actual times for the MRI investigations turned out to be close to those intended, that is, 4, 10, 20, and 42 days after injury; the mean times (±1SD, range) being 3.9 (±0.7, 2-5) days, 10.2 (±0.8, 9-12) days, 20.3 (±1.6, 18-24) days, and 42.8 (±2.8, 40-50) days, respectively.

No avulsion injuries or complete muscle or tendon ruptures were present and the radiographs of the pelvis were normal in all the sprinters. The long head of the biceps femoris muscle constituted the primary injury site in all 18 sprinters. Eight (44%) of the sprinters had a secondary injury, 7 in the semitendinosus and 1 in the short head of the biceps femoris. There was total agreement between the 2 radiologists about the locations of the primary and secondary injuries in all 18 sprinters.

The location of the primary injury defined in relation to the different regions of the muscle-tendon complex of the long head of the biceps femoris, as defined in Figure 1, is presented for each of the 18 sprinters in Table 2. Proximal muscle-tendon junction (PMTJ) and/or proximal tendon (PT) were involved in 12 cases, and distal muscle-tendon junction (DMTJ), distal tendon (DT), and/or distal muscle-belly

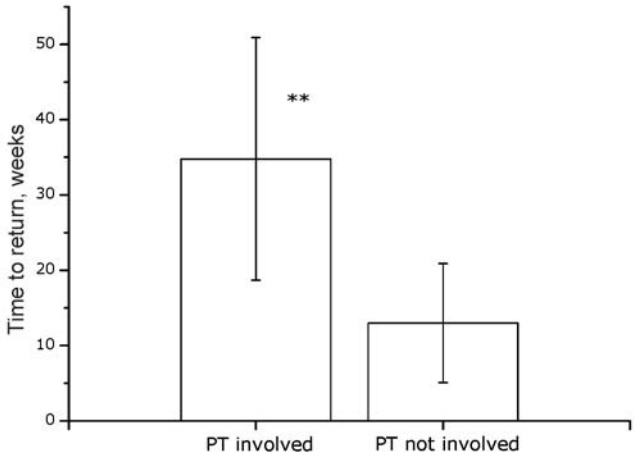


**Figure 2.** Time to return to pre-injury level for the sprinters with injuries either involving the proximal free tendon (PT) of the biceps femoris long head (n = 6) or not involving the PT (n = 12). Values are means ±1 SD. **denotes a significant difference between the 2 groups (*P* = .009, Mann-Whitney *U* test).

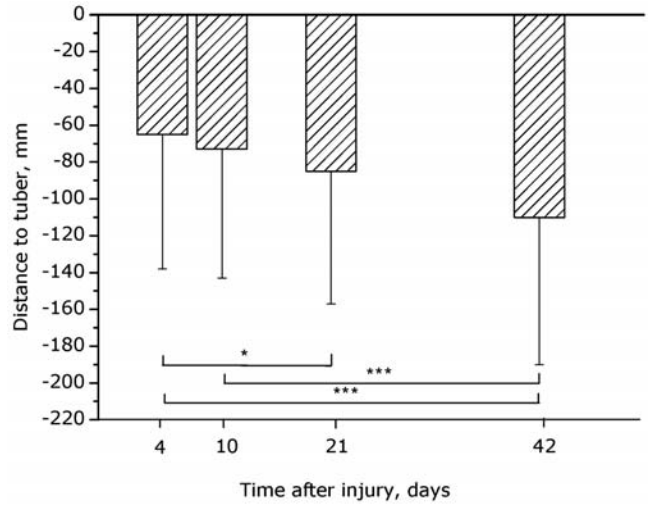


**Figure 3.** Distance from the most cranial pole of the injury to the most caudal part of the ischial tuberosity at the 4 MRI investigations (I-IV). Negative values indicate that the injury is distal to the ischial tuberosity. Values are means ±1 SD. Significant differences between MRI investigations (I-IV) are indicated. **P* < .05; ***P* < .001.

TABLE 3
Mean Values ±1 SD (and Ranges) for MRI Parameters at the 4 MRI Investigations (I-IV)

| Parameter | MRI I (n = 17) | MRI II (n = 17) | MRI III (n = 15) | MRI IV (n = 18) |
|---|---|---|---|---|
| Distance to tuber, mm | 67 ± 71 (–21-218) | 79 ± 65 (0-205) | 86 ± 69 (6-232) | 102 ± 79 (17-240) |
| Length, mm | 187 ± 74 (60-346) | 159 ± 64 (39-296) | 144 ± 81 (33-287) | 90 ± 60 (0-229) |
| Width, mm | 22 ± 10 (8-50) | 18 ± 6 (6-28) | 18 ± 8 (7-32) | 12 ± 6 (0-22) |
| Depth, mm | 33 ± 17 (8-66) | 27 ± 14 (7-57) | 25 ± 12 (9-48) | 19 ± 12 (0-44) |
| Volume, $cm^3$ | 88 ± 101 (4-414) | 44 ± 38 (8-109) | 40 ± 45 (1-161) | 17 ± 19 (0-67) |
| Cross-sectional area, % | 38 ± 23 (10-90) | 32 ± 19 (7-69) | 26 ± 16 (5-50) | 17 ± 15 (0-50) |

(DMB) in 8. The subgroup where the injury involved PT (n = 6) had a significantly longer time back to pre-injury level compared with the subgroup with no PT involvement (n = 12) (P = .009) (Figure 2). The subgroup in which the injury involved proximal muscle-tendon junction (PMTJ) + PT (n = 5) had a tendency toward longer time back to pre-injury level than the subgroup with only PMTJ involvement (n = 5) (P = .064). A comparison between the subgroup where the PMTJ and/or PT were involved (n = 12) with the subgroup where distal muscle-tendon junction (DMTJ) and/or distal tendon/distal muscle-belly (DT/DMB) were involved (n = 6) showed a significantly longer time back to pre-injury level for the former, proximally injured, subgroup (P = .028).

There was a gradual increase in the distance from the most cranial pole of the injury to the ischial tuberosity with time after injury (Table 3, Figure 3). A similar change was seen for the disappearance of the injury as evidenced by gradual decreases of its length, width, depth, volume, and cross-sectional area from MRI investigation I to IV, as illustrated in Figure 4 and quantified in Table 3 and Figure 5. At the last MRI investigation, 6 weeks after the injury, the primary injury was completely healed, that is, no signs of the injury appeared on MRI, in only 1 of the 18 sprinters. Correspondingly, 6 (75%) of the 8 secondary injuries had disappeared at MRI investigation IV. The average size of the injury remaining after 6 weeks, as indicated by the measured MRI parameters, ranged from 20% to 55% of the values on the first test occasion (Figure 5).

A clear majority of the calculated correlations (19 of 24) at the 4 MRI investigations demonstrated a significantly positive relationship between the various quantitative measures of the extent of the injury and the time to return to pre-injury level (Table 4). Correlation analysis also showed that the shorter the distance to tuber was from the most cranial pole of the injury at each of the 4 test occasions, the longer the time was to return to pre-injury level (Table 4).

## DISCUSSION

### Prediction of Recovery Time

One major new finding of the present study was that the injury location in terms of distance from the ischial tuberosity

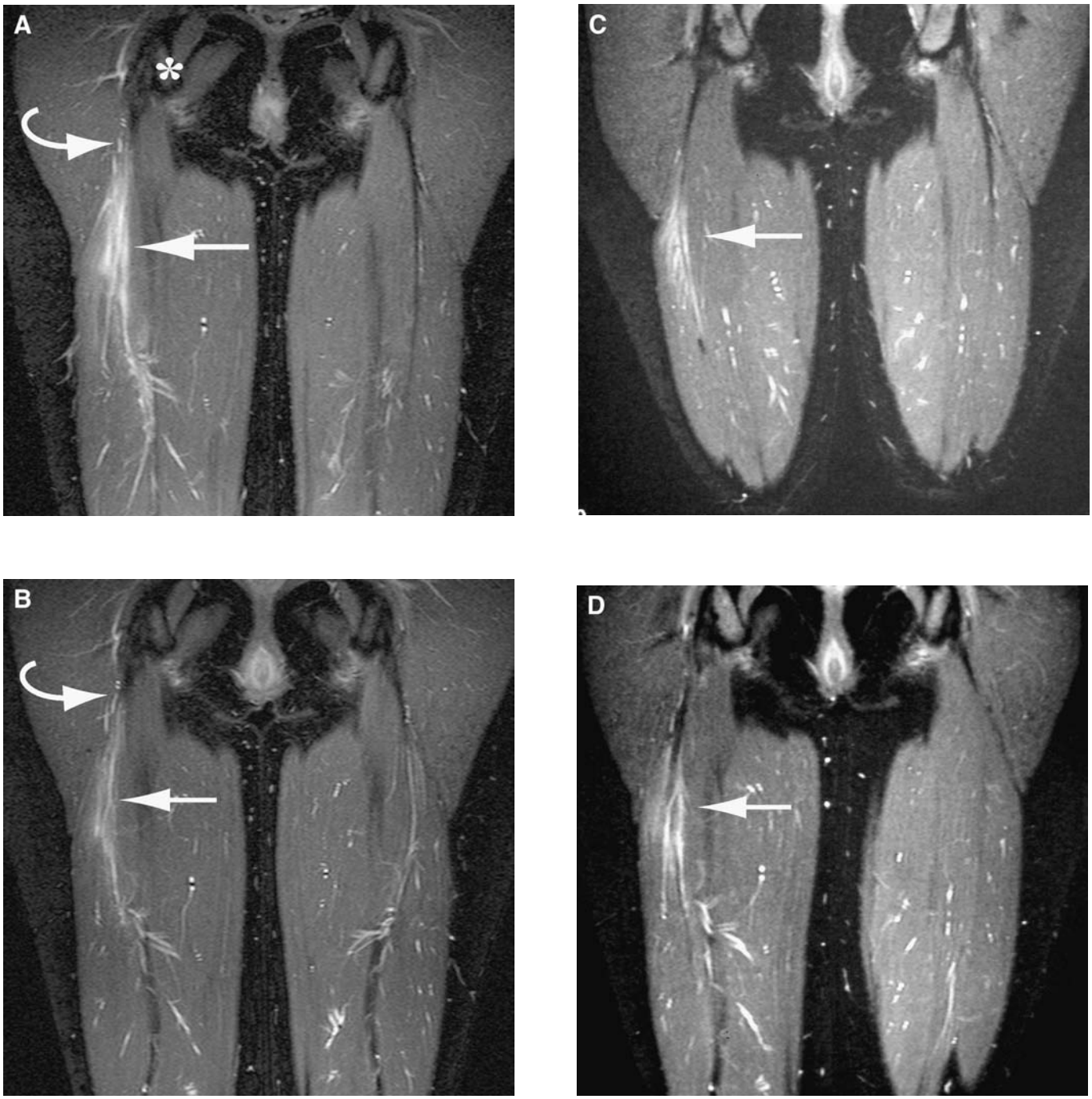


**Figure 4.** The progression of a typical hamstring injury in 1 sprinter. MRIs (A-D) were obtained on days 4, 10, 21, and 42 after injury, respectively. The injury was located in the long head of the biceps femoris, initially (A) involving the proximal muscle-tendon junction (PMTJ, straight arrow) and the proximal tendon (PT, curved arrow). *indicates the location of the ischial tuberosity. The sequence with the largest edema was used in each MRI investigation. There was a gradual decrease in size of the injury over time. It took this athlete 25 weeks to return to pre-injury level. The values for selected MRI parameters at occasions I-IV (A-D) were for length, 193, 161, 121, and 107 mm, and for distance from the most cranial pole of the injury to the ischial tuberosity, 11, 23, 52, and 50 mm. The distance from the location of the highest pain on palpation to the ischial tuberosity was 7, 3, 5, and 8 cm, respectively.

was associated with recovery time: the more cranial the injury, the longer the time back. Interestingly, the prediction of recovery time was equally good using the point of highest pain on palpation, established within 3 weeks of the injury occasion, as using the distance from the most cranial pole of the injury determined with MRI. This indicates that

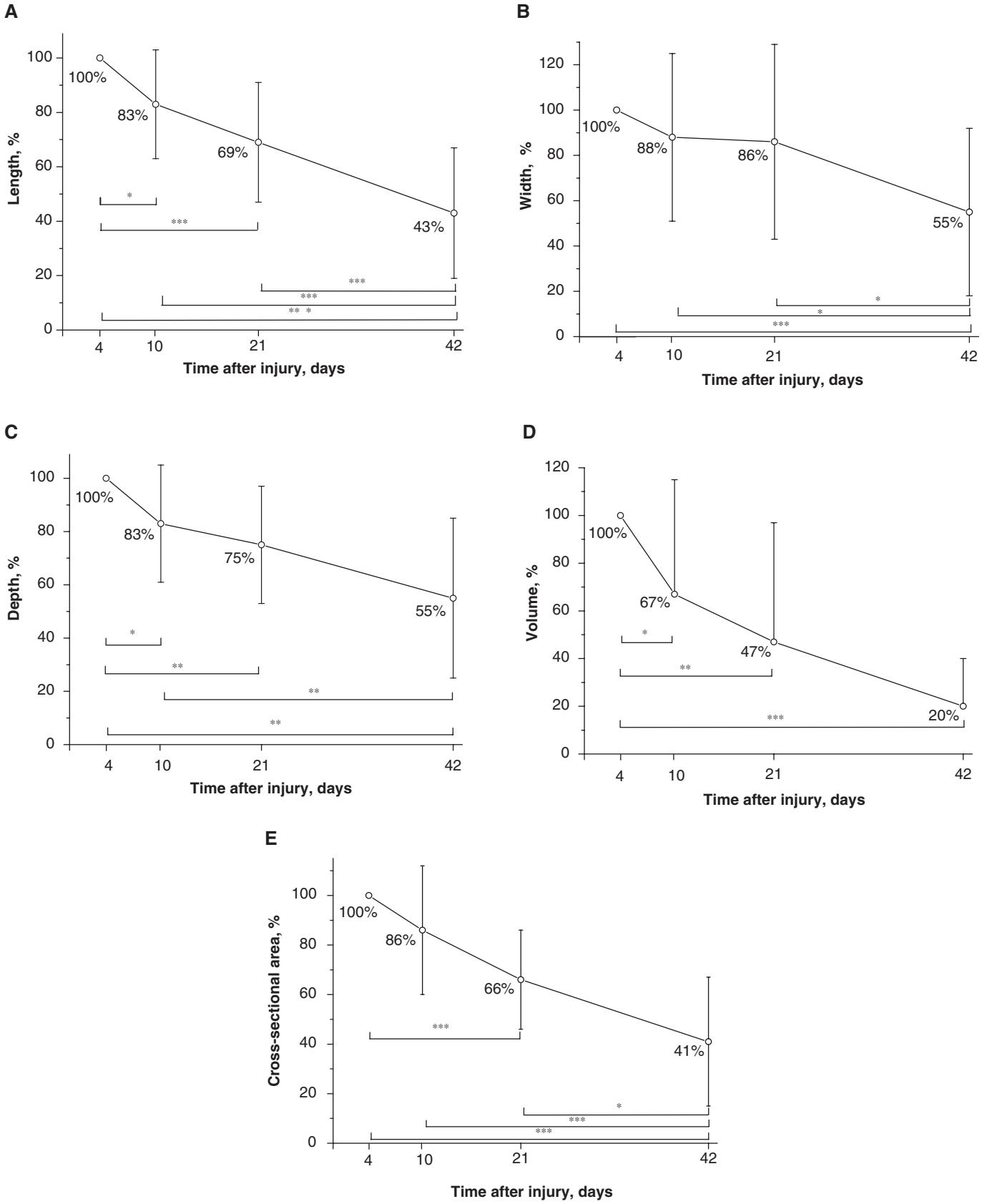


**Figure 5.** Changes in MRI parameters over time (MRI investigations I-IV) expressed as the extent of the remaining injury in percentage of the corresponding values at MRI investigation I. Values are means ±1 SD. Significant differences between MRI investigations (I-IV) are indicated. *$P < .05$; **$P < .01$; ***$P < .001$.

TABLE 4
Correlations (r) Between MRI Parameters at Each of the 4 MRI Investigations (I-IV) and Time to Return to Pre-injury Level[a]

| | MRI I (n = 17) | | MRI II (n = 17) | | MRI III (n = 15) | | MRI IV (n = 18) | |
|---|---|---|---|---|---|---|---|---|
| MRI Parameters | r | *P* | r | *P* | r | *P* | r | *P* |
| Distance to tuber | 0.544[b] | 0.044 | 0.599[b] | 0.023 | 0.691[c] | 0.006 | 0.705[c] | 0.005 |
| Length | 0.505[d] | 0.055 | 0.372 | 0.172 | 0.582[b] | 0.023 | 0.726[c] | 0.002 |
| Width | 0.394 | 0.146 | 0.384 | 0.158 | 0.423 | 0.116 | 0.186 | 0.507 |
| Depth | 0.584[b] | 0.022 | 0.573[b] | 0.025 | 0.662[c] | 0.007 | 0.607[b] | 0.017 |
| Volume | 0.608[b] | 0.016 | 0.537[b] | 0.039 | 0.716[c] | 0.003 | 0.653[c] | 0.008 |
| Cross-sectional area | 0.695[c] | 0.004 | 0.636[b] | 0.011 | 0.533[b] | 0.041 | 0.463 | 0.082 |

[a]Before calculation the times were log transformed to achieve normal distribution.
[b]$P < .05$.
[c]$P < .01$.
[d]$.05 \leq P < .1$.

repeated, carefully performed clinical examinations during the first 3 weeks after injury can give important information about the prognosis of a hamstring strain. Furthermore, correlations existed at all MRI investigations, up to 6 weeks, which indicates that MRI is valuable for estimating recovery time even at a rather late stage of rehabilitation.

The other major new finding of the present study was that defining the region of injury within the muscle-tendon complex of the long head of biceps femoris was of significance for the prognosis of recovery time. Involvement of the free proximal tendon was associated with longer time to return to pre-injury level. This is in accordance with the results from the palpation, even though palpation evidently cannot distinguish between tissues involved. No previous study on hamstring strains has systematically compared the consequences of involvement of free tendon (cf. Figure 1) versus muscle/muscle-tendon junction on recovery time. Our earlier case study indicated, however, that free tendon involvement might prolong the recovery time.[2] It can also be noted that injuries affecting free tendons in general (eg, Achilles, patellae, and supraspinatus tendons) are usually accompanied by prolonged rehabilitation times.

Our findings of good prediction of recovery time for the sprinters from the size and/or the position of the injury in relation to the ischial tuberosity (more or less cranial) as well as proximal tendon involvement are illustrated by the 2 extreme cases within our material shown in Figure 6 (data on each individual are presented in the figure legend). For the sprinter with the shortest time back (6 weeks) (Figure 6, A and B), the size of the initial injury was small and the distance to the ischial tuberosity was long compared to the sprinter with the longest recovery time (50 weeks) (Figure 6, C and D). The latter had an injury involving the proximal free tendon, whereas the former did not. Interestingly, these 2 sprinters showed very similar initial symptoms, both fell dramatically during their races, and both had to use crutches for 4 days after injury. Also, isometric knee flexion strength and range of motion in hip flexion with straight knee showed a similar decrement at the first clinical examination in the 2 sprinters (remaining strength of the uninjured leg, 47% vs 53%; range of motion, 80% vs 64%, respectively). At the last test, 6 weeks after injury, the values for the 2 sprinters were as follows: remaining strength, 87% vs 101%; range of motion, 96% versus 70%, respectively. The sprinter with the shortest time back was healed at the last MRI investigation, but the sprinter with the longest time back still had 64% left of the length of the injury at the last MRI investigation.

## Specificity of the Material and the Injuries

This study was carried out on a well-defined injury in a specific group of athletes. Strict inclusion criteria resulted in a rather small group of subjects, albeit a 3-year study period, but, on the other hand, in a homogeneous group of athletes with acute, first-time hamstring strain injuries of similar origin. All the sprinters included in the study after the initial clinical examination had their hamstring injury verified by MRI. Careful history ensured that the injury was a true first-time injury, which disqualified a majority of potential subjects. All subjects had a specific injury mechanism, namely straightforward, undisturbed, competitive running at maximal or close to maximal speed. Thus the conditions were well controlled with respect to running surface, shoes, fatigue, and warm-up. Although this study was limited to sprinters, it should be possible to extrapolate the results to other individual sports, as well as to team sports, where hamstring injuries are considered to occur frequently at high running speeds.

All primary injuries were located in the long head of the biceps femoris. This result in sprinters is in line with previous studies showing that this is the most commonly injured hamstring muscle in high-force/high-speed sports.[4,5,12,17,18] Our observation that 7 of the 8 secondary injuries occurred in the semitendinosus muscle is also in accordance with earlier studies on other types of athletes reporting a combined injury of biceps femoris and semitendinosus as the most common "tandem injury."[5,11] The reason for the predominance of hamstring strain injuries in these particular parts of the hamstring muscle synergy is not known. Anecdotal evidence and biomechanical modeling of high-speed running suggest that the long head of the biceps femoris should be the muscle

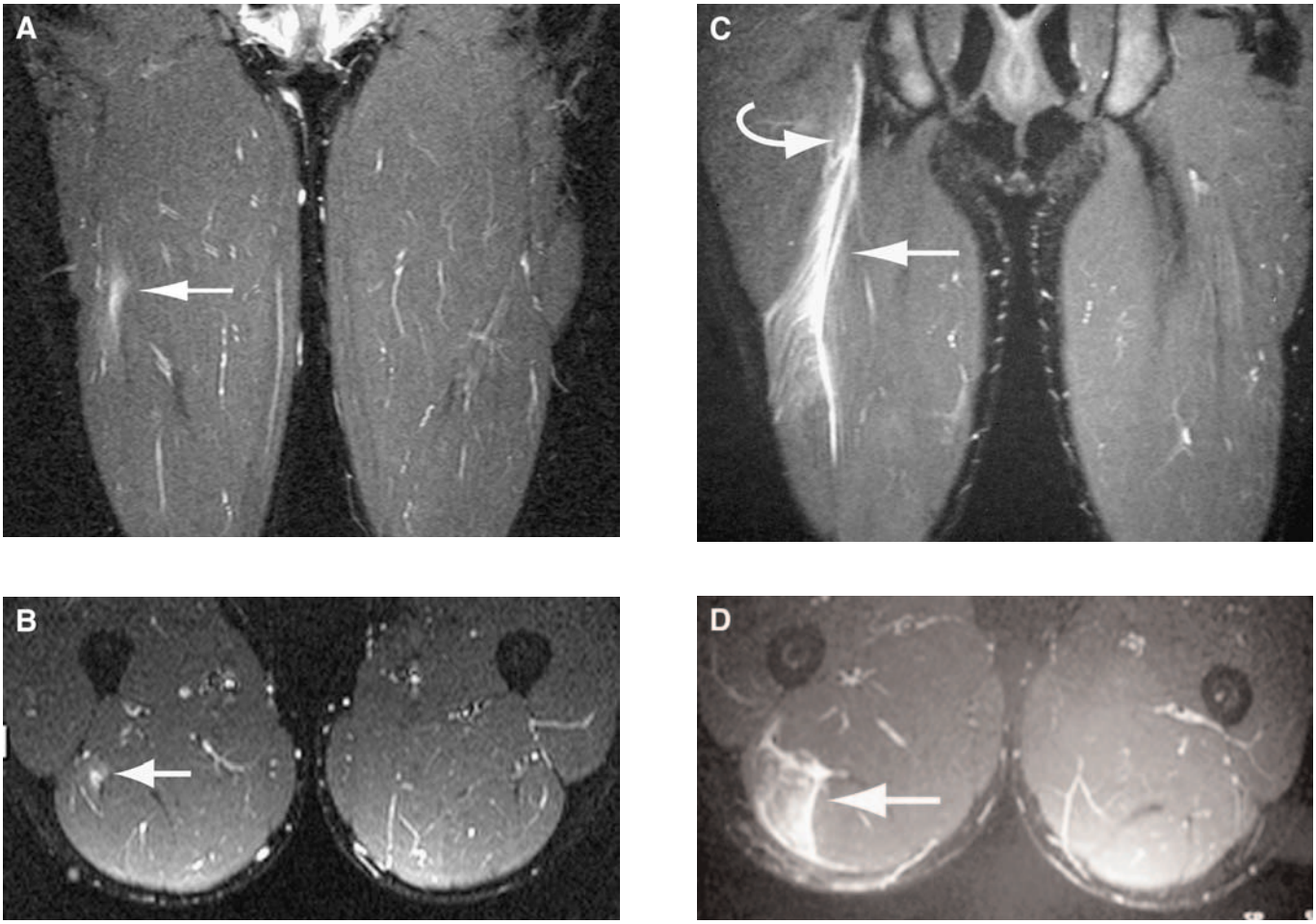


**Figure 6.** Frontal (A, C) and transverse (B, D) MRI images (STIR) from the first MRI investigation 4 days after injury; first (A, B), for the sprinter with the shortest time (6 weeks) and, second (C, D), the sprinter with the longest time to return to pre-injury level (50 weeks). In both, the injury was located in the long head of the biceps femoris, involving, in the first sprinter (A), only a part of the PMTJ (arrow) and in the second sprinter (C), PMTJ (straight arrow) and PT (curved arrow). The values for selected MRI parameters in the first and second sprinters were as follows: distance from the most cranial pole of the injury to the ischial tuberosity, 107 mm distal and 21 mm proximal; length, 60 and 251 mm; volume, 42 and 414 $cm^3$; and cross-sectional area, 15% and 90%, respectively. The distance from the location of the highest pain on palpation to the ischial tuberosity was 16 and 5 cm caudal to tuber, respectively.

undergoing the most lengthening when activated at the end of the swing phase and/or early support phase in fast running.[9] But, for obvious reasons, systematic data on the actual injury mechanism are lacking.

Although various methods have been applied to classify which regions and tissues in the muscle tendon complex are injured,[5,11,17] most studies agree that the muscle-tendon junction is involved.[5,17] Of importance in this context is to acknowledge the complex anatomy of the hamstring muscles with well-defined free tendons and extensive intramuscular tendons,[7] as highlighted in the detailed anatomical study by Woodley and Mercer.[19] The present study shows that the free tendons also are affected by the injury.

## Progression of the Injury: Status After 6 Weeks

The acute effects of the injury were conspicuous. At the first clinical examination, 2 days after injury, 15 of the 18 sprinters used crutches, and isometric knee flexor strength and range of motion in hip flexion with straight knee were markedly reduced.[1] Remaining strength and flexibility amounted to 38% and 61%, respectively, of that of the uninjured leg. However, the recovery was remarkably fast. Within 10 days after the injury, none of the sprinters needed crutches during walking, and strength and flexibility were back to more than 70% of the values for the uninjured leg. After 6 weeks, the performance of the injured leg in strength and flexibility tests exceeded 90% of that of the uninjured leg, but only 2 sprinters considered themselves ready to run at maximal speed. Thus, a 90% to 95% level of test performance, as often is recommended as a criterion for full return to sports, was not in agreement with the sprinters' own judgment that they could return safely to maximal speed running.

Interestingly, the MRI investigation at the same occasion, 6 weeks after injury, showed remaining signs of injury in all

but 1 of the sprinters. That sprinter was 1 of the 2 who were able to run on their pre-injury level at this time. On the average, 20% to 55% of the extent of the injury remained 6 weeks after injury and, as mentioned, all the sprinters, except 2, were still not back at the pre-injury level of performance. A recent study on Australian footballers[4] similarly demonstrated remains of MRI abnormalities at 6 weeks after injury. However, in their study, all players but 1 had returned to play at that time. Also in other studies on team sports,[3] it has been reported that players recovered far quicker than the sprinters in our study. It could be speculated that a reason for this might be that athletes in team sports can return to playing matches without being entirely back to their pre-injury performance level, whereas sprinters probably need much closer to full function.[13]

No ecchymosis was observed in any of the clinical examinations and, accordingly, the MRI investigations showed that the injuries were mainly located intrafascially. All the sprinters experienced pain on palpation at 3 weeks, but one third of them were pain-free at 6 weeks. Notably, only 1 of these 6 pain-free cases completely lacked MRI indications of injury at this time. In general, there was good agreement between the results of palpation and the MRI findings. The palpation results showed consistency in locating the point of highest pain, and the MRI analysis demonstrated that this point fell within the longitudinal extent of the injury in 85% of all cases at all 4 test occasions. The point of highest pain was always located in the lateral part of the rear thigh, which is consistent with the MRI signs of injury primarily situated in the long head of the biceps femoris muscle.

One of the limitations of our study relates to control of the rehabilitation during the follow-up period. After the first 6 weeks, the rehabilitation of the subjects was trusted with their own physicians and physical therapists, and therefore there was a possibility of varying methods being employed. However, all subjects in this study came from a similar high competition level, having daily access to sport physicians, sports physical therapists, and training coaches. These rehabilitation teams have corresponding levels of education and training and thus are likely to apply similar treatment and rehabilitation programs for hamstring strains. Thereby, the potential influence on the current results should be minimized.

## CONCLUSION

Careful palpation during the first 3 weeks after injury and MRI investigation during the first 6 weeks after injury provide useful information about time to return to a pre-injury level of performance in elite sprinting.

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